

PAPER_697: NGC 2525 with SN 2018gv: Type Ia Supernova Light Curve and UQFF

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Abstract

NGC 2525 is a barred spiral galaxy (SBc) at 60 Mly hosting Type Ia SN 2018gv, observed by Hubble in time-lapse. The Arnett light curve model and Phillips relation are applied with UQFF modifications to the peak luminosity.

1. Parameters

Parameter	Value
M_{galaxy}	$3 \times 10^{10} M_{\odot}$
Distance	60 Mly
$L_{SN,peak}$	1.5×10^{43} W
M_{Ni56}	$0.6 M_{\odot}$
Rise time	14 days

2. Type Ia Light Curve (Arnett)

$$L(t) \approx L_{peak} e^{-\frac{1}{2} \left(\frac{t-t_r}{\sigma_t} \right)^2} \times e^{-(t-t_r)/t_d}$$

3. Phillips Relation

$$M_B \approx -19.3 + 0.74(\Delta m_{15} - 1.1)$$

Δm_{15} : magnitude decline in first 15 days post-peak.

4. UQFF Luminosity Correction

$$L_{UQFF}(t) = L_{SN}(t) \left(1 + \frac{\rho_{SCm}}{\rho_{UA}} \right) (1 - f_{TRZ})$$

5. Barred Disk Rotation

$$v_c(R) = \sqrt{v_{disk}^2 + v_{bar}^2}$$

where $v_{bar}^2 = GM_{gal} \cdot 0.3(1 - e^{-R/R_{bar}})/R$.

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- Arnett (1982), ApJ, 253, 785
- Phillips (1993), ApJ, 413, L105
- UQFF Framework, Star-Magic Session 174

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